Posterior Inference, Prediction, and Decisions

An undergraduate introduction to Bayesian risk analysis

Based on *Bayes Rules!*, Chapter 8

Source: https://www.bayesrulesbook.com/chapter-8

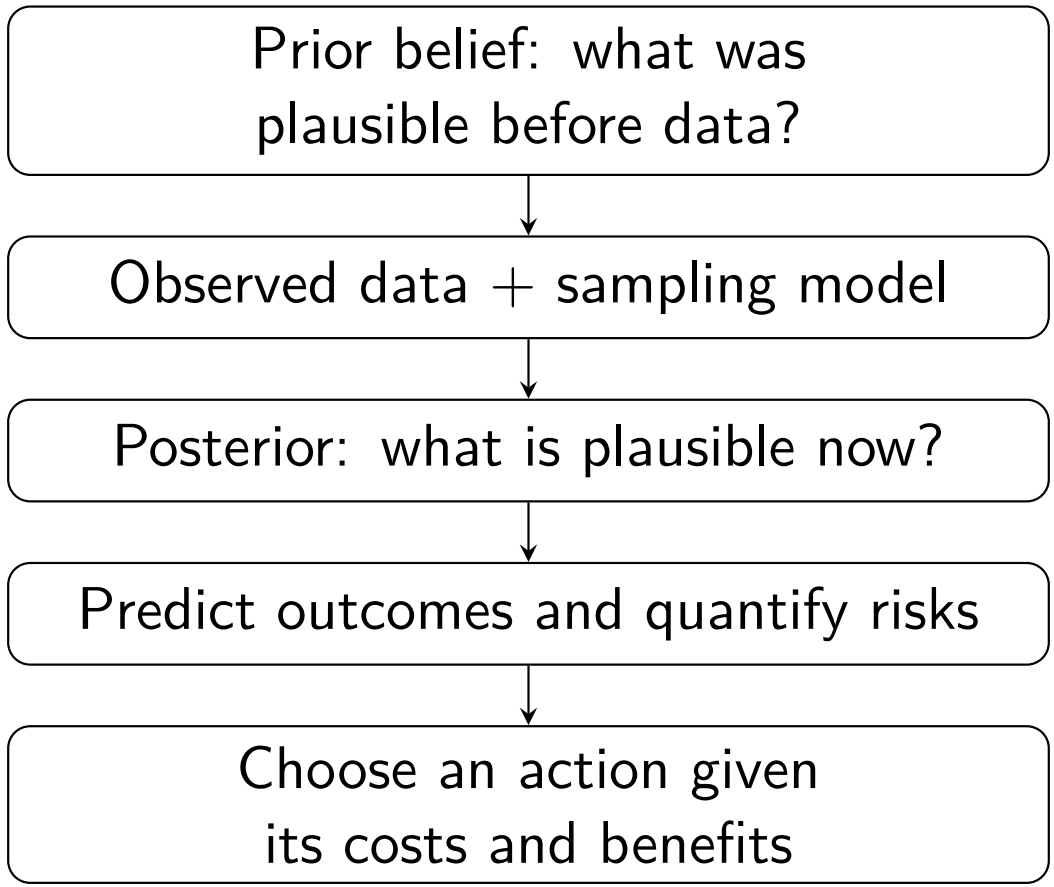
Learning goals

By the end, you should be able to:

1. Summarize a posterior using a mean and credible interval.
2. Calculate the probability of a decision-relevant claim.
3. Predict future observations, including uncertainty.
4. Compare actions using **expected loss** and a risk tolerance.
5. Explain how simulation supports these calculations.

Big idea: Bayesian analysis does not end with an estimate. It helps us *choose an action under uncertainty*.

The Bayesian workflow



Running example from Chapter 8: museum artists

Let π be the proportion of artists in major US modern-art museums who were born in 1965 or later.

- Prior: $\pi \sim \text{Beta}(4, 6)$.
- Sample: 14 of 100 artists in a MoMA sample are in this group.
- Sampling model: $Y \mid \pi \sim \text{Binomial}(100, \pi)$.

$$\boxed{\pi \mid Y = 14 \sim \text{Beta}(4 + 14,\ 6 + 86) = \text{Beta}(18, 92)}$$

Interpretation: after seeing the data, smaller proportions are more plausible than the prior suggested.

A posterior is more than one number

For $\pi \mid Y = 14 \sim \text{Beta}(18, 92)$:

- Posterior mean: $\mathbb{E}(\pi \mid Y) = 18/110 \approx \mathbf{0.164}$.
- Posterior mode: $(18 - 1)/(110 - 2) \approx 0.157$.
- Middle 95% credible interval: $(\mathbf{0.101}, \mathbf{0.238})$.

Plain language: the proportion is likely near 16%, but values around 10% to 24% remain plausible. **Risk lesson:** two estimates with the same mean can lead to different decisions if their uncertainty differs.

Understanding credible intervals

A 95% Bayesian credible interval $[L, U]$ is a range satisfying

$$\mathbb{P}(L \leq \pi \leq U \mid Y) = 0.95.$$

For our example:

$$\mathbb{P}(0.101 \lesssim \pi \lesssim 0.238 \mid Y = 14) \approx 0.95.$$

Do not confuse two questions

- **Parameter uncertainty:** What proportion π is plausible?
- **Future variability:** What might happen in the *next* sample?

A credible interval addresses the first question, not the second.

Posterior probabilities answer useful questions

Suppose a planning team asks:

Is the true proportion below 20%?

For the museum posterior,

$$\mathbb{P}(\pi < 0.20 \mid Y = 14) = \mathbf{0.849}.$$

- The model assigns about 85% probability to the claim.
- The complementary probability is about 15%.
- This is *not* a guarantee; choices also depend on consequences.

Decision connection: a threshold matters when it changes what we would do.

Hypotheses and Bayes factors: the intuition

Compare two claims:

$$H_a : \pi < 0.20, \qquad H_0 : \pi \geq 0.20.$$

- Prior chance of H_a: 0.0856.
- Posterior chance of H_a: 0.849.
- Posterior odds: $0.849/0.151 \approx 5.62$.

$$\text{Bayes factor} = \frac{\text{posterior odds}}{\text{prior odds}} \approx \mathbf{60}.$$

The data shifted the odds strongly toward H_a. **Important:** a large Bayes factor is evidence, *not* a decision rule; costs of mistakes still matter.

A caution about exact-value hypotheses

For a continuous parameter with a continuous posterior,

$$\mathbb{P}(\pi = 0.30 \mid Y) = 0.$$

That does *not* automatically establish a useful decision.

- Instead, ask whether a value is **practically different**.
- Example: "roughly 30%" means $0.25 \leq \pi \leq 0.35$.
- Calculate posterior probability of this *range*.

Decision lesson: define a meaningful tolerance before looking at results.

Prediction: what happens next?

Suppose we examine 20 *new* museum artists; Y' is the number born in 1965 or later.

$$Y' \mid \pi \sim \text{Binomial}(20, \pi).$$

The expected future count is

$$\mathbb{E}(Y' \mid Y = 14) = 20\mathbb{E}(\pi \mid Y = 14) = 20(18/110) \approx \mathbf{3.27}.$$

But observing exactly 3 is not certain! Prediction includes **sampling randomness** and **uncertainty about** π.

Posterior predictive distribution

To predict a future count, average over plausible values of π:

$$\mathbb{P}(Y' = k \mid Y) = \int_0^1 \mathbb{P}(Y' = k \mid \pi)\, p(\pi \mid Y)\, d\pi.$$

In the Beta–Binomial model this gives a **Beta–Binomial** predictive distribution.

- Predicted mean number of younger artists out of 20: 3.27.
- Chance of at least 5: $\mathbb{P}(Y' \geq 5 \mid Y = 14) \approx \mathbf{0.233}$.

Risk lesson: often the most important answer is a *tail probability*, not the expected count.

From inference to decision analysis

A decision requires **three ingredients**:

1. **Actions:** What can we choose?
2. **Uncertain states/outcomes:** What could happen?
3. **Loss or payoff:** What is the consequence of each action and state?

For an action a, define its posterior expected loss:

$$\boxed{R(a \mid Y) = \mathbb{E}\big[L(a, \theta) \mid Y\big].}$$

Choose the action with the **smallest expected loss** if that is the agreed decision objective.

New example: quality-control risk

A manufacturer wants to estimate θ, the probability that a new part fails inspection.

- Prior: $\theta \sim \text{Beta}(2, 18)$ (mean 10%).
- Inspect 32 parts; 3 fail and 29 pass.
- Posterior: $\theta \mid Y \sim \text{Beta}(5, 47)$.

$$\mathbb{E}(\theta \mid Y) = \frac{5}{52} \approx \mathbf{9.6\%}.$$

95% credible interval: approximately **3.3% to 18.9%**.
Management question: Should we perform an extra screening step before shipment?

Define the available actions and losses

Suppose the team can either **ship as is** or **screen every part**. For teaching purposes, assume:

- Shipping an unchecked failed part costs \$200 on average.
- Screening costs \$12 per part.
- Screening identifies and resolves every failure at no further cost.

For a future part,

$$\mathbb{E}(L_{\text{ship}} \mid Y) = 200\mathbb{E}(\theta \mid Y),$$
$$\mathbb{E}(L_{\text{screen}} \mid Y) = 12.$$

Note: These are simplified, illustrative cost assumptions, not real factory estimates.

Compute expected loss and choose

Under the posterior mean failure rate $5/52$:

$$R(\text{ship} \mid Y) = 200 \times \frac{5}{52} \approx \$19.23,$$
$$R(\text{screen} \mid Y) = \$12.00.$$

Decision under the stated assumptions

Screen the part: lower posterior expected loss by about **$7.23 per part**.

For 1,000 comparable future parts, the expected saving is about $7,230. This result depends on both the posterior *and* the assumed costs.

Break-even thresholds are easy to communicate

Screening is preferable when

$$12 < 200\mathbb{E}(\theta \mid Y).$$

Equivalently,

$$\boxed{\mathbb{E}(\theta \mid Y) > 0.06.}$$

- If expected failure probability exceeds 6%, screen.
- Our estimate is 9.6%, so screen.
- If costs change, the **break-even threshold changes**.

Sensitivity question: what if screening costs \$22 rather than \$12?

Probability of crossing a risk threshold

The posterior can also answer:

$$\mathbb{P}(\theta > 0.10 \mid Y) = \mathbb{P}(\theta > 0.10 \mid \text{Beta}(5, 47)) \approx \mathbf{0.413}.$$

There is about a **41%** chance the failure rate exceeds 10%.

- This probability could trigger further inspection or monitoring.
- It need not change the expected-loss decision automatically.
- Different organizations may have different risk tolerances.

Prediction for a batch: another form of risk

Suppose 20 more parts will be shipped.

$$K \mid \theta \sim \text{Binomial}(20, \theta), \qquad \theta \mid Y \sim \text{Beta}(5, 47).$$

Using the posterior predictive distribution:

$$\mathbb{E}(K \mid Y) = 20(5/52) \approx \mathbf{1.92},$$

$$\mathbb{P}(K \geq 3 \mid Y) \approx \mathbf{0.306}.$$

Thus the expected count is about 2, but there is still about a **31% chance of 3 or more** failures.

Expected loss vs. downside risk

Both may matter, but they answer different questions.

Criterion	**Question**
Expected loss	Which action has the smallest average cost?
Tail probability	How likely is an unacceptably bad outcome?
Credible interval	How uncertain is the underlying rate?
Predictive distribution	How variable might future results be?

A robust decision report makes the **objective, threshold, and assumptions** explicit.

What is the value of more information?

Imagine we could inspect more parts before deciding on screening.

- New data might push the estimated failure rate *above or below* the 6% break-even point.
- That could change the preferred action.
- Collecting more data has a cost, so we should ask whether it might *improve the decision enough* to justify that cost.

Core idea: information is valuable when it can change an action and reduce expected loss.

Simulation: inference when formulas are difficult

In a simple Beta–Binomial model, we can calculate many answers exactly. In more complex models we often simulate from the posterior (e.g., MCMC).

1. Draw many plausible $\theta^{(1)}, \ldots, \theta^{(S)}$ values.
2. Approximate $\mathbb{E}(\theta \mid Y)$ by their average.
3. Approximate $\mathbb{P}(\theta > 0.10 \mid Y)$ by the fraction over 0.10.
4. For each draw, simulate future failures $K^{(s)}$.
5. Approximate risks and expected losses using simulated outcomes.

Practical check: MCMC results need convergence and sampling diagnostics.

Optional R: four questions in a few lines

```
# Museum example, Chapter 8
18 / (18 + 92)                    # posterior mean
qbeta(c(.025, .975), 18, 92)  # credible interval
pbeta(.20, 18, 92)                # P(pi < .20 | data)

# Quality-control example
qbeta(c(.025, .975), 5, 47)
pbeta(.10, 5, 47, lower.tail = FALSE)
200 * 5 / 52                      # expected loss: ship
12                                # expected loss: screen
```

For the predictive tail, use the self-contained code on the next slide (no extra packages).

Optional R: posterior predictive tail

```
# Beta-Binomial probability, using base R
pbb <- function(k, n, a, b) {
  exp(lchoose(n, k) + lbeta(a + k, b + n - k)
      - lbeta(a, b))
}

# Museum: P(at least 5 out of the next 20)
sum(sapply(5:20, pbb, n = 20, a = 18, b = 92))

# Parts: P(at least 3 failures out of the next 20)
sum(sapply(3:20, pbb, n = 20, a = 5, b = 47))
```

Interpretation: add probabilities of all outcomes that meet the adverse-event threshold.

Class exercise: would you screen?

A new product has posterior failure probability

$$\theta \mid Y \sim \text{Beta}(3, 27), \qquad \mathbb{E}(\theta \mid Y) = 0.10.$$

Suppose inspection costs \$15 per unit, and an unchecked failure costs \$100.

1. What is the expected loss of shipping without inspection?
2. What is the expected loss of inspecting every unit?
3. Which action minimizes expected loss?
4. If an unchecked failure cost \$250, would your answer change?

Class exercise: discussion answers

1. Ship: $100 \times 0.10 = \$\mathbf{10}$ expected loss.
2. Inspect: $\$\mathbf{15}$ expected loss.
3. With a \$100 failure cost: **ship**, under the stated assumptions.
4. With a \$250 failure cost: ship costs $250 \times 0.10 = \$25$; **inspect** is preferred.

Takeaway: the data and posterior are unchanged; the optimal action changes because the consequences change.

Five questions to ask in any risk analysis

1. What is uncertain, and what is the posterior distribution?
2. What threshold or future event actually matters?
3. Which actions are available?
4. What are the costs and benefits of each outcome?
5. Would additional information be worth collecting?

Posterior $\longrightarrow$ **prediction** $\longrightarrow$ **loss** $\longrightarrow$ **decision**